MCE5919: Gaussian Process for Machine Learning and Signal Processing

Lecture 1

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Shenzhen Research Institute of Big Data & School of Science and Engineering
The Chinese University of Hong Kong, Shenzhen

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Overview

1. Self Introduction

2. About This Course

3. Introduction to Machine Learning

4. Motivation of GP

5. Application

Self Introduction

Instructor

- Dr. Wenqiang Pu, Research Building B, Room 307
- Email: wenqiangpu@cuhk.edu.cn/wpu@sribd.cn

Education

- 2009-2013, BSc., Xidian University, China
- 2013-2018, Ph.D., Xidian University, China
- 2015-2018, Visiting Ph.D., CUHKSZ, China

Work Experience

- 2019-2020, Postdoc, CUHKSZ, China
- 2020-present, Research Scientist, SRIBD, China
- 2024-present, Adjunct Assistant Professor, CUHKSZ, China

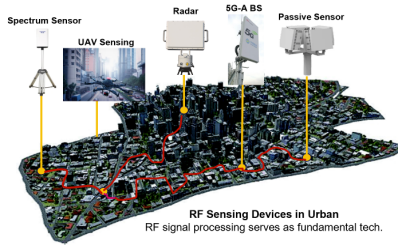
Self Introduction

Research Interest

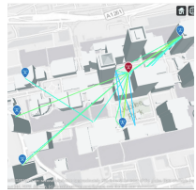
- Wireless Signal Processing
- Optimization Algorithms
- Machine Learning for SP Applications

Open Intern Positions at SRIBD

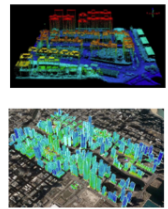
- Multimodal Sensing for UAVs
- Auto SP Algorithm Discovery
- Environment-aware Signal Processing



RF Sensing Devices in Urban
RF signal processing serves as fundamental tech.



Ray Tracking



LiDAR Point Cloud

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About This Course

Lecture

Time and Venue

- Time: Monday 15:30-17:00, Wednesday 19:00-20:30
- Lecture Room: TA302 (Monday), TxC402 (Wednesday)
- Teaching Assistant: Mr. Chao Hu
- Office Hour: Wednesday 15:00-17:00, Research Building B, Room 307

Preliminaries

- Math: Linear Algebra, Statistics, Optimization
- Programming: Python, MATLAB

About This Course

Textbooks

- Gaussian Processes for Machine Learning[1]
- Machine Learning: A Bayesian and Optimization Perspective[2]

[1] Williams C. K., Rasmussen C E. Gaussian Processes for Machine Learning. Cambridge, MA: MIT Press, 2006.

[2] Theodoridis S. Machine Learning: a Bayesian and Optimization Perspective. Academic Press, 2015.

About This Course

Learning Objective

- To understand fundamental knowledge in Machine Learning and Bayesian Learning
- To get familiar with Gaussian Process and techniques in Bayesian Learning
- To know a wide range of applications of Gaussian Process
- To grasp the general routine of applying learning methods to real problems

Grading Scheme

- Assignment (30/100, 3 assignments throughout the semester)
- Team Project (30/100, 2-3 students one team, 30 min presentation)
- Final Exam (40/100, at the end of the semester)

About This Course

Lecture Topics

- Provably Approximately Correct Learning
- Linear and Logistic Regression
- Kernel Methods and Gaussian Process
- GP for Regression and Classification
- Learning in Reproducing Kernel Hilbert Space
- Optimal Kernel Design
- Deep Kernel Learning
- Applications

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What Machine Learning is About

Learn from Data through Proper Models

- Learning through personal experience and knowledge that propagates from generation to generation is at the heart of human intelligence
- The heart of any scientific fields is the development of models in order to explain the available experimental evidence

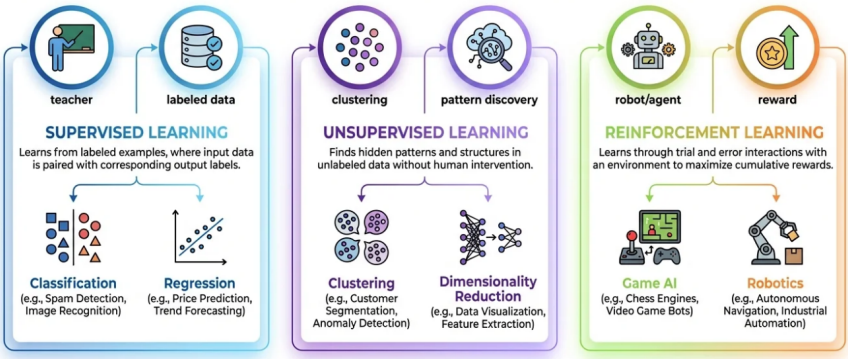




What Machine Learning is About

Types of Learning

- Supervised v.s. Unsupervised Learning
- Active v.s. Passive Learning
- Online v.s. Batch Learning
- ...



Statistical Learning

- In this course, we focus on statistical learning framework

$$\min_{h \in \mathcal{H}} \mathbb{E}_{z \sim \mathcal{D}} [\ell(h, z)] := L_{\mathcal{D}}(h)$$

  - z – data sample, $z = (\mathbf{x}, y)$ with $\mathbf{x}$ being the feature vector and y being the label
  - $\mathcal{D}$ – probability distribution over domain $\mathcal{X} \times \mathcal{Y}$ ($\mathbf{x} \in \mathcal{X}, y \in \mathcal{Y}$)
  - $\mathcal{H}$ – hypothesis class, e.g., finite set, linear model, neural network
- Choice of loss function

$$\text{0-1 loss:} \quad \ell_{0-1}(h, (x, y)) \stackrel{\text{def}}{=} \begin{cases} 0 & \text{if } h(x) = y \\ 1 & \text{if } h(x) \neq y \end{cases}$$

$$\text{square loss:} \quad \ell_{\text{sq}}(h, (x, y)) \stackrel{\text{def}}{=} (h(x) - y)^2$$

Statistical Learning

- For a finite data sample set $\mathcal{S} = \{z_1, z_2, \ldots, z_n\}$, $z_i \sim \mathcal{D}$, i.i.d.
- The empirical risk minimization (ERM) is

$$\min_{h \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^{n} \ell(h, z_i) := L_{\mathcal{S}}(h)$$

Is ERM a good learning criterion?[3]

[3] Optional lecture on PAC learning

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Bayes Inference

- So far, the learning models we consider are deterministic but unknown
- Next, we treat our attention to a different rationale
- The unknown parameters are random variable
- This corresponds to estimate value of a specific realization from the observed data
- The heart of this direction beats around the Bayes Theorem

Bayes Inference

- Given two jointly distributed random vector $\mathbf{y}$ and $\boldsymbol{\theta}$
- Bayes Theorem states that

$$p(\mathbf{y}, \boldsymbol{\theta}) = p(\mathbf{y} \mid \boldsymbol{\theta}) p(\boldsymbol{\theta}) = p(\boldsymbol{\theta} \mid \mathbf{y}) p(\mathbf{y})$$

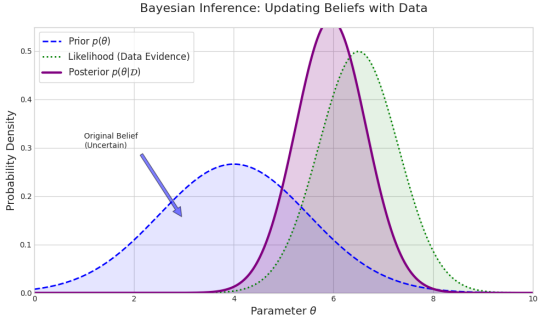
- Let $\mathcal{D} = \{(\mathbf{x}_i, y_i) \in \mathbb{R}^d \times \mathbb{R}, i = 1, 2, \ldots, n\}$ be the set of observation
- Then similarly we have

$$p(\boldsymbol{\theta} \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \boldsymbol{\theta}) p(\boldsymbol{\theta})}{p(\mathcal{D})} = \frac{p(\mathcal{D} \mid \boldsymbol{\theta}) p(\boldsymbol{\theta})}{\int p(\mathcal{D} \mid \boldsymbol{\theta}) p(\boldsymbol{\theta}) d\boldsymbol{\theta}}$$

- The random vector $\boldsymbol{\theta}$ can be treated as a random variable

Bayes Inference

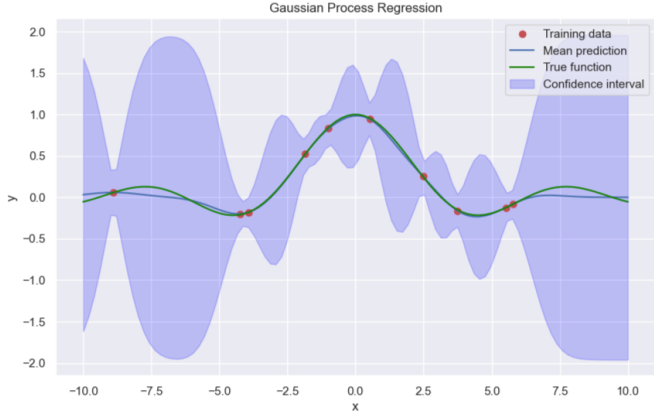
- $p(\boldsymbol{\theta})$ is the prior distribution that encapsulates our original knowledge and belief about the unknowns $\boldsymbol{\theta}$ before collecting any evidence
- $p(\boldsymbol{\theta} \mid \mathcal{D})$ is the posterior distribution, i.e., the refined belief, formed after the set of n observations has been obtained
- The process of approximating the pdf of a random quantity based on a set of training data is referred as inference



Motivation of GP

- Gaussian Process can be 'regarded' as Bayes inference for function

$$f(\mathbf{x}) \sim \mathcal{GP}(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}'))$$



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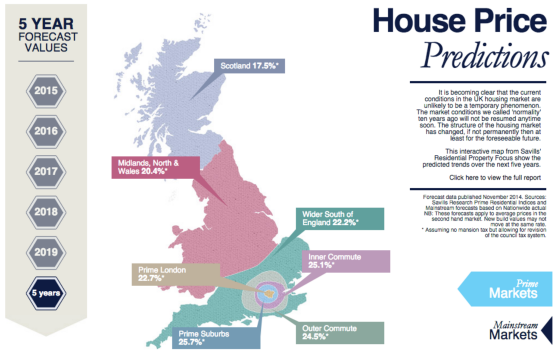
4. Motivation of GP

5. **Application**

Application

Example-I: House price prediction in UK

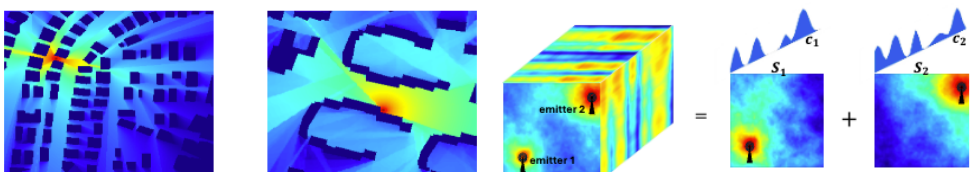
Input: $\mathbf{x} = (\text{time}, \text{position})$, Output: $y = \text{House price at a specific time.}$



Application

Example-II: Radio Map reconstruction

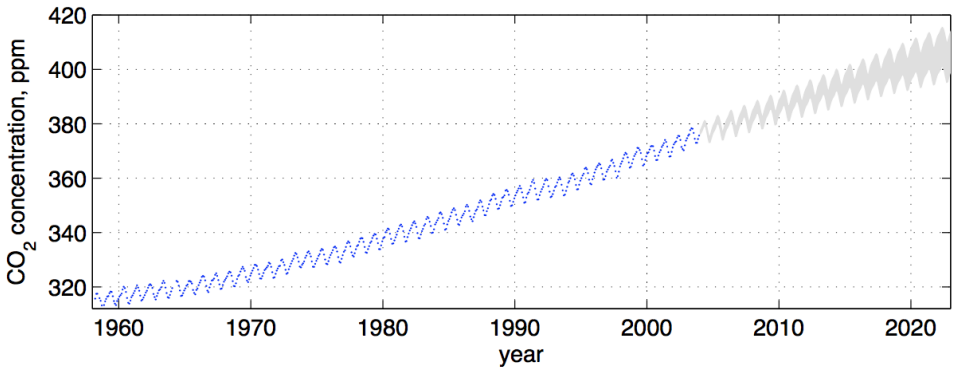
Input: two-dimensional position $\mathbf{x} = (p_1, p_2)$, Output: $y = \mathrm{RSS\ value}$.



Application

Example-III: Prediction of monthly averages of the atmospheric concentration of CO2.

Input: x =time, Output: y =CO2 concentration (unit in ppm).



Application

Example-IV: Handwritten digits recognition.

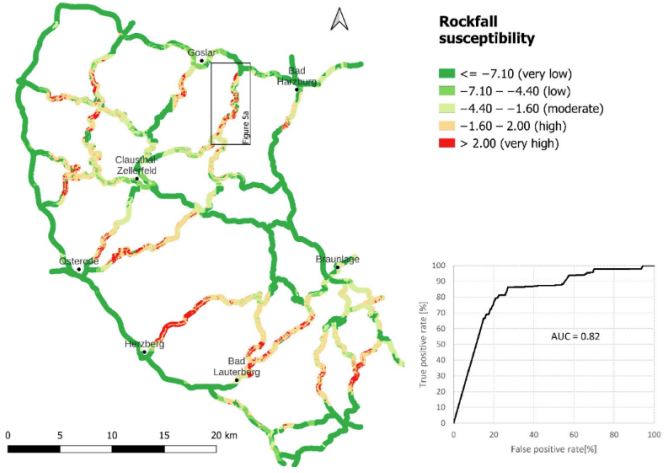
Input: x= (stacked) image pixels, Output: y =class index $\{1, 2, ..., 9\}$.



Application

Example-V: Traffic Routing Game.

Input: x= selected routine of each player, Output: y = total time cost.



The End